Comparing Alternative Parameterizations

Of the Dirichlet-Multinomial Distribution

For Fitting Composition Data

Jon Brodziak, NOAA Fisheries, Pacific Islands Fisheries Science Center

Jon.Brodziak@NOAA.GOV

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We outline an analytical framework to evaluate how alternative forms of the Dirichlet-multinomial distribution compare for fitting composition data. Specifically, we examine how two parameterizations of the Dirichlet-multinomial (DM) distribution, a linear form and a saturated form, compare with the standard unconstrained parameterization in fitting simulated categorical data. We also compare the DM model with the multinomial model that assumes independent trials with fixed probabilities for each category and often underrepresents variability in real-world compositional data due to heterogeneity and clustering. In constrast, the DM models category probabilities as random draws from a Dirichlet distribution, which captures both overdispersion and inter-category correlation. Our objective is to assess which DM parameterization performs best for fitting composition data across a range of sample size and dispersion scenarios.

We begin with the standard DM and introduce two DM parameterizations as described by Thorson et al. (2017) and Fisch et al. (2021). We begin with a glossary of key terms and symbols, that describe the data, parameters, and likelihoods needed to estimate proportion vectors from composition data (Table 1). This provides a consistent basis for the simulation and estimation approaches to compare the accuracy of the three DM forms.

Composition data are counts or proportions of individuals classified into predefined categories—such as age or size classes—capturing the distribution of characteristics within a statistical sampling frame. We will use age-structured data as the type of composition data sampled from the fish stock. The number of categories, or age classes, is denoted by A, with the age bins a=1, 2, …, (A-1) representing true ages and the and the age bin A representing the plus group consisting of all individuals A and older.

Three types of count vectors are used in this study. The vector of actual numbers at age in the stock is $\mathbf{x}$. The observed count vector from the sampling frame is $\mathbf{x}_{obs}$ with sample size n_{obs}. The predicted count vectors estimated by optimizing the parameters of the multinomial or DM likelihood fitted to the observed data is $\mathbf{x}_{pred}$.

Table 1. Glossary of terms.

Variable	Description
A	Number of categories of age or size bins indexed by a = 1, 2, …, A
$\mathbf{x}$	True population count vector with elements x_a
$\mathbf{x}_{obs}$	Observed count vector with elements $\mathrm{x}_{obs,a}$
$\mathbf{x}_{pred}$	Predicted count vector with elements $\mathrm{x}_{pred,a}$
$\mathbf{p}$	True population proportion vector with elements p_a
$\mathbf{p}_{obs}$	Observed proportion vector with elements $\mathrm{p}_{obs,a}$
$\mathbf{p}_{pred}$	Predicted proportion vector with elements $\mathrm{p}_{pred,a}$
n_{obs}	Observed composition sample size
$\boldsymbol{\alpha}$	Concentration parameter vector of a Dirichlet-multinomial distribution
α_0	Sum of concentration parameters of a Dirichlet-multinomial distribution
θ	Weight parameter of a linear Dirichlet-multinomial distribution
β	Weight parameter of a saturated Dirichlet-multinomial distribution
n_{eff}	Effective sample size of a observed proportion vector

Three types of proportion vectors are used in this study. The first is the true population proportion vector at age in the stock is $\mathbf{p}$. The observed proportion vector is $\mathbf{p}_{obs}$. The predicted proportion vector estimated from the observed data is $\mathbf{p}_{pred}$. We deterministically simulate the true population proportion vector $\mathbf{p}$ and sample the population proportion vector $\mathbf{p}_{obs}$ with observation error. We then estimate the population proportion vector $\mathbf{p}_{pred}$ by optimizing the parameters of the multinomial or DM likelihoods. The simulated population $\mathbf{p}$ could also include process error due to random fluctuations in demographic factors and environmental conditions. The sampled population $\mathbf{p}_{obs}$ will include observation error due to sample size and random variation in observed samples as well as coverage and vulnerability to the sampling gear. The predicted population $\mathbf{p}_{pred}$ includes uncertainty due to estimation error and the amount of contrast and variability in the observed data input to the composition likelihoods.

The multinomial distribution models the probabilities of counts across multiple categories when a fixed number of independent trials each result in a single categorical outcome, making it useful for estimating proportions from age composition data. It assumes that each trial is independent, that the category probabilities remain constant across trials, and that outcomes are mutually exclusive, typically requiring a simple random sampling design from a well-defined population. Proportions are estimated by dividing the observed counts by the total number of trials, yielding maximum likelihood estimates of the true category probabilities. The multinomial likelihood for an observed set of counts $\mathbf{x}_{obs} = \mathrm{n}_{obs} \cdot \mathbf{p}_{obs}$ is

(1.1)
$$L_{MN}\left(\mathbf{p}_{pred} \mid \mathbf{x}_{obs}\right) = \frac{\Gamma\left(\mathrm{n}_{obs}+1\right)}{\prod_{a=1}^{A}\Gamma\left(\mathrm{x}_{obs,a}+1\right)} \prod_{a=1}^{A}\left(\mathrm{p}_{pred,a}\right)^{\mathrm{n}_{obs}\cdot \mathrm{p}_{obs,a}}$$

and the multinomial negative loglikelihood NLL_{MN} for a set of observed composition data is

(1.2)
$$NLL_{MN}\left(\mathbf{p}_{pred} \mid \mathbf{x}_{obs}\right) = -\mathrm{n}_{obs}\sum_{a=1}^{A}\mathrm{p}_{obs,a}\cdot\log\left(\mathrm{p}_{pred,a}\right) - \log\left(\mathrm{n}_{obs}+1\right) + \sum_{a=1}^{A}\log\left(\mathrm{x}_{obs,a}+1\right)$$

In equation (1.2) the parameters of the NLL_{MN} are in the vector of predicted proportions $\mathbf{p}_{pred} = \left(\mathrm{p}_{pred,1},\ldots,\mathrm{p}_{pred,A}\right)$ at time *t* as a function of the input sample size n_{obs} and the observed proportions $\mathbf{p}_{obs} = \left(\mathrm{p}_{obs,1},\ldots,\mathrm{p}_{obs,A}\right)$.

The multinomial negative loglikelihood NLL_{MN} for composition data sampled in a set of *T* time periods by a set of *V* fishing fleets indexed by *v* is

(1.3)

$$NLL_{MN}\left(\mathbf{P}_{pred} \mid \mathbf{n}_{obs}, \mathbf{P}_{obs}\right) = -\sum_{v=1}^{V}\sum_{t=1}^{T}\left[\mathrm{n}_{obs,v,t}\sum_{a=1}^{A}\mathrm{p}_{obs,v,t,a}\cdot\log\left(\mathrm{p}_{pred,t,a}\right) - \log\left(\mathrm{n}_{obs}+1\right) + \sum_{a=1}^{A}\log\left(\mathrm{x}_{obs,a}+1\right)\right]$$

where the capitalized boldface $\mathbf{P}_{obs}$ is a *VxTxA*-dimensional array and $\mathbf{P}_{pred}$ is a *TxA*-dimensional array with proportion vectors at age in the last dimension. The parameters of the NLL_{MN} are the 2-D array of predicted proportions $\mathbf{P}_{pred} = \left[\mathrm{p}_{pred,t,a} \right]_{TxA}$ at time *t* as a function of the input sample sizes $\mathbf{n}_{obs} = \left[\mathrm{n}_{obs,v,t} \right]_{VxT}$ and the array of observed proportions $\mathbf{P}_{obs} = \left[\mathrm{p}_{obs,v,t,a} \right]_{VxTxA}$.

The Dirichlet-multinomial distribution extends the multinomial distribution by allowing the category probabilities to vary according to a Dirichlet distribution, making it suitable for modeling overdispersed count data. Like the multinomial, it models counts across multiple categories from a fixed number of trials, but it assumes that the underlying category probabilities are not fixed but randomly drawn from a Dirichlet prior, reflecting variability across samples or clusters. This distribution accounts for extra variation due to unobserved heterogeneity or correlated observations, such as in cluster sampling, where the assumption of independent trials is violated. Proportions are still estimated from observed counts, but the Dirichlet-multinomial model inflates the variance of those estimates to better reflect real-world sampling variability.

The standard Dirichlet-multinomial distribution is a function of its unconstrained concentration parameter vector $\boldsymbol{\alpha}$ and the observed set of counts $\mathbf{x}_{obs}$, where the $\alpha_j > 0$ are positive and sum to α_0 and $\mathbf{x}_{obs} = \mathrm{n}_{obs} \cdot \mathbf{p}_{obs}$. Samples from the distribution will be more clumped or overdispersed for lower α_0 and will be more similar to multinomial behavior for larger α_0, noting that the unconstrained Dirichlet-multinomial distribution approaches the multinomial as $\alpha_0 \to \infty$. The likelihood of the unconstrained Dirichlet-multinomial distribution is

$$L_{DMU}\left(\boldsymbol{\alpha} \mid \mathbf{x}_{obs}\right) = \frac{\Gamma\left(\mathrm{n}_{obs}+1\right)}{\prod_{a=1}^{A}\Gamma\left(\mathrm{x}_{obs,a}+1\right)} \frac{\Gamma\left(\alpha_0\right)}{\Gamma\left(\mathrm{n}_{obs}+\alpha_0\right)} \prod_{a=1}^{A} \frac{\Gamma\left(\mathrm{x}_{obs,a}+\alpha_a\right)}{\Gamma\left(\alpha_a\right)} \tag{1.4}$$

and the standard unconstrained Dirichlet-multinomial negative loglikelihood NLL_{DMU} is

$$\begin{aligned} NLL_{DMS}\left(\boldsymbol{\alpha} \mid \mathbf{x}_{obs}\right) = &-\log\left(\Gamma\left(\mathrm{n}_{obs}+1\right)\right) + \sum_{a=1}^{A}\log\left(\Gamma\left(\mathrm{x}_{obs,a}+1\right)\right) \\ &-\log\left(\Gamma\left(\alpha_0\right)\right) + \log\left(\Gamma\left(\mathrm{n}_{obs}+\alpha_0\right)\right) \\ &-\sum_{a=1}^{A}\log\left(\Gamma\left(\mathrm{x}_{obs,a}+\alpha_a\right)\right) + \sum_{a=1}^{A}\log\left(\Gamma\left(\alpha_a\right)\right) \end{aligned} \tag{1.5}$$

The partial derivative of the unconstrained Dirichlet-multinomial negative loglikelihood with respect to the concentration parameter α_a is

$$(1.6) \qquad \frac{\partial NLL_{DMU}\left(\boldsymbol{\alpha} \mid \mathbf{x}_{obs}\right)}{\partial \alpha_a} = -\Psi\left(\alpha_0\right) + \Psi\left(\mathrm{n}_{obs} + \alpha_0\right) - \Psi\left(\mathrm{x}_{obs,a} + \alpha_a\right) + \Psi\left(\alpha_a\right)$$

where Ψ is the digamma function or derivative of the log-gamma function with $\Psi = \frac{\Gamma'}{\Gamma}$.

We want to contrast the estimation performance of the standard form and two alternative forms of the Dirichlet-multinomial distribution used for fitting composition data. The first form is the saturated Dirichlet-multinomial distribution (DMS). The DMS likelihood for a single period is

$$(1.7) \qquad L_{DMS}\left(\mathbf{p}_{pred}, \beta \mid \mathbf{x}_{obs}\right) = \frac{\Gamma\left(\mathrm{n}_{obs}+1\right)}{\prod_{a=1}^{A}\Gamma\left(\mathrm{n}_{obs}\mathrm{p}_{obs,a}+1\right)} \frac{\Gamma\left(\beta\right)}{\Gamma\left(\mathrm{n}_{obs}+\beta\right)} \prod_{a=1}^{A} \frac{\Gamma\left(\mathrm{n}_{obs}\mathrm{p}_{obs,a} + \beta \cdot \mathrm{p}_{pred,a}\right)}{\Gamma\left(\beta \cdot \mathrm{p}_{pred,a}\right)}$$

where the parameters are the vector of predicted proportions $\mathbf{p}_{pred} = \boldsymbol{\alpha} / \alpha_0$ and the weight parameter $\beta = \alpha_0$. The DMS distribution differs from the DMU distribution for numerical optimization because it includes the parameter constraint that the predicted proportions must sum to unity. The saturated Dirichlet-multinomial negative loglikelihood NLL_{DMS} is

$$(1.8) \qquad \begin{aligned} NLL_{DMS}\left(\mathbf{p}_{pred}, \beta \mid \mathbf{x}_{obs}\right) = & -\log\left(\Gamma\left(\mathrm{n}_{obs}+1\right)\right) + \sum_{a=1}^{A} \log\left(\Gamma\left(\mathrm{n}_{obs}\mathrm{p}_{obs,a}+1\right)\right) \\ & -\log\left(\Gamma\left(\beta\right)\right) + \log\left(\Gamma\left(\mathrm{n}_{obs}+\beta\right)\right) \\ & -\sum_{a=1}^{A} \log\left(\Gamma\left(\mathrm{n}_{obs}\mathrm{p}_{obs,a} + \beta\mathrm{p}_{pred,a}\right)\right) + \sum_{a=1}^{A} \log\left(\Gamma\left(\beta\mathrm{p}_{pred,a}\right)\right) \end{aligned}$$

The partial derivatives of the saturated Dirichlet-multinomial negative loglikelihood with respect to the predicted proportion $\mathrm{p}_{pred,a}$ and the weight parameter β are

$$(1.9) \qquad \frac{\partial NLL_{DMS}\left(\mathbf{p}_{pred}, \beta \mid \mathbf{x}_{obs}\right)}{\partial \mathrm{p}_{pred,a}} = -\Psi\left(\mathrm{n}_{obs}\mathrm{p}_{obs,a} + \beta\mathrm{p}_{pred,a}\right) + \Psi\left(\beta\mathrm{p}_{pred,a}\right)$$

and

(1.10)
$$\frac{\partial NLL_{DMS}\left(\mathbf{p}_{pred},\beta \mid \mathbf{x}_{obs}\right)}{\partial \beta}=-\Psi\left(\beta\right)+\Psi\left(\mathrm{n}_{obs}+\beta\right)-\sum_{a=1}^{A}\left[\Psi\left(\mathrm{n}_{obs}\mathrm{p}_{obs,a}+\beta\cdot\mathrm{p}_{pred,a}\right)-\Psi\left(\beta\mathrm{p}_{pred,a}\right)\right]$$

The saturated Dirichlet-multinomial negative loglikelihood for a set of T sampling periods and a set of V fishing fleets with fleet-specific weight parameters is

(1.11)

$$NLL_{DMS}\left(\mathbf{P}_{pred},\beta|\mathbf{n},\mathbf{P}_{obs}\right)=-\sum_{v=1}^{V}\sum_{t=1}^{T}\left\{\begin{array}{l}\log\left(\Gamma\left(\mathrm{n}_{obs,v,t}+1\right)\right)-\sum_{a=1}^{A}\log\left(\Gamma\left(\mathrm{n}_{obs,v,t}\mathrm{p}_{obs,v,t,a}+1\right)\right)\\ +\log\left(\Gamma\left(\beta_{v}\right)\right)-\log\left(\Gamma\left(\mathrm{n}_{obs,v,t}+\beta_{v}\right)\right)\\ +\sum_{a=1}^{A}\log\left(\Gamma\left(\mathrm{n}_{obs,v,t}\mathrm{p}_{obs,v,t,a}+\beta_{v}\mathrm{p}_{pred,t,a}\right)\right)-\sum_{k=1}^{K}\log\left(\Gamma\left(\beta_{v}\mathrm{p}_{pred,t,a}\right)\right)\end{array}\right\}$$

The parameters of the NLL_{DMS} are the predicted proportions $\mathbf{P}_{pred}$ by period and the fleet-specific weight parameters β_v as a function of $\mathbf{n}_{obs}$ and the observed proportions $\mathbf{P}_{obs}$.

The second form is the linear Dirichlet-multinomial negative loglikelihood NLL_{DML} which is

(1.12)
$$\begin{aligned}NLL_{DML}\left(\mathbf{p}_{pred},\theta|\mathrm{n}_{obs},\mathbf{p}_{obs}\right)=&-\log\left(\Gamma\left(\mathrm{n}_{obs}+1\right)\right)+\sum_{a=1}^{A}\log\left(\Gamma\left(\mathrm{n}_{obs}\mathrm{p}_{obs,a}+1\right)\right)\\&-\log\left(\Gamma\left(\theta\mathrm{n}_{obs}\right)\right)+\log\left(\Gamma\left(\mathrm{n}_{obs}+\theta n_{t}\right)\right)\\&-\sum_{a=1}^{A}\log\left(\Gamma\left(\mathrm{n}_{obs}\mathrm{p}_{obs,a}+\theta\mathrm{n}_{obs}\mathrm{p}_{pred,a}\right)\right)+\sum_{k=1}^{K}\log\left(\Gamma\left(\theta\mathrm{n}_{obs}\mathrm{p}_{pred,a}\right)\right)\end{aligned}$$

where $\theta=\beta/\mathrm{n}_{obs}$ is the linear Dirichlet-multinomial weight parameter.

The partial derivatives of the saturated Dirichlet-multinomial negative loglikelihood with respect to the predicted proportion $\mathrm{p}_{pred,a}$ and the weight parameter θ are

(1.13)
$$\frac{\partial NLL_{DML}\left(\mathbf{p}_{pred},\theta \mid \mathbf{x}_{obs}\right)}{\partial \mathrm{p}_{pred,a}}=-\Psi\left(\mathrm{n}_{obs}\cdot\mathrm{p}_{obs,a}+\theta\mathrm{n}_{obs}\mathrm{p}_{pred,a}\right)+\Psi\left(\theta\mathrm{n}_{obs}\mathrm{p}_{pred,a}\right)$$

and

$$\frac{\partial NLL_{DML}\left(\mathbf{p}_{pred},\theta \mid \mathbf{x}_{obs}\right)}{\partial\theta} = -\Psi\left(\theta \mathrm{n}_{obs}\right)+\Psi\left(\mathrm{n}_{obs}+\theta \mathrm{n}_{obs}\right) - \sum_{a=1}^{A}\left[\Psi\left(\mathrm{n}_{obs}\mathrm{p}_{obs,a}+\theta \mathrm{n}_{obs}\mathrm{p}_{pred,a}\right)-\Psi\left(\theta \mathrm{n}_{obs}\mathrm{p}_{pred,a}\right)\right] \quad (1.14)$$

The linear Dirichlet-multinomial negative loglikelihood for a set of T sampling periods and a set of V fishing fleets with equal weight parameters is

$$NLL_{DML}\left(\mathbf{P}_{pred},\theta \mid \mathbf{n}_{obs},\mathbf{P}_{obs}\right) = -\sum_{v=1}^{V}\sum_{t=1}^{T}\left\{\begin{array}{l} \log\left(\Gamma\left(\mathrm{n}_{obs,v,t}+1\right)\right)-\sum_{a=1}^{A}\log\left(\Gamma\left(\mathrm{n}_{obs,v,t}\cdot \mathrm{p}_{obs,v,t,a}+1\right)\right) \\ \log\left(\Gamma\left(\theta \mathrm{n}_{obs,v,t}\right)\right)-\log\left(\Gamma\left(\mathrm{n}_{obs,v,t}+\theta \mathrm{n}_{obs,v,t}\right)\right) \\ \sum_{a=1}^{A}\log\left(\Gamma\left(\mathrm{n}_{obs,v,t}\cdot \mathrm{p}_{obs,v,t,a}+\theta \mathrm{n}_{obs,v,t}\cdot \mathrm{p}_{pred,v,t,a}\right)\right) \\ -\sum_{a=1}^{A}\log\left(\Gamma\left(\theta \mathrm{n}_{obs,v,t}\cdot \mathrm{p}_{pred,v,t,a}\right)\right) \end{array}\right\} \quad (1.15)$$

The parameters of the NLL_{DML} are the predicted proportions $\mathbf{P}_{pred}$ by period and the weight parameter θ as a function of $\mathbf{n}_{obs}$ and the observed proportions $\mathbf{P}_{obs}$.

References

Fisch, N., Camp, E., Shertzer, K., Ahrens, R. 2021. Assessing likelihoods for fitting composition data within stock assessments, with emphasis on different degrees of process and observation error. Fisheries Research, 243, 106069, https://doi.org/10.1016/j.fishres.2021.106069

Thorson, J.T., Johnson, K.F., Methot, R.D., Taylor, I.G. 2017. Model-based estimates of effective sample size in stock assessment models using the Dirichlet-multinomial distribution. Fisheries Research, 192:84-93, https://doi.org/10.1016/j.fishres.2016.06.005

Appendix

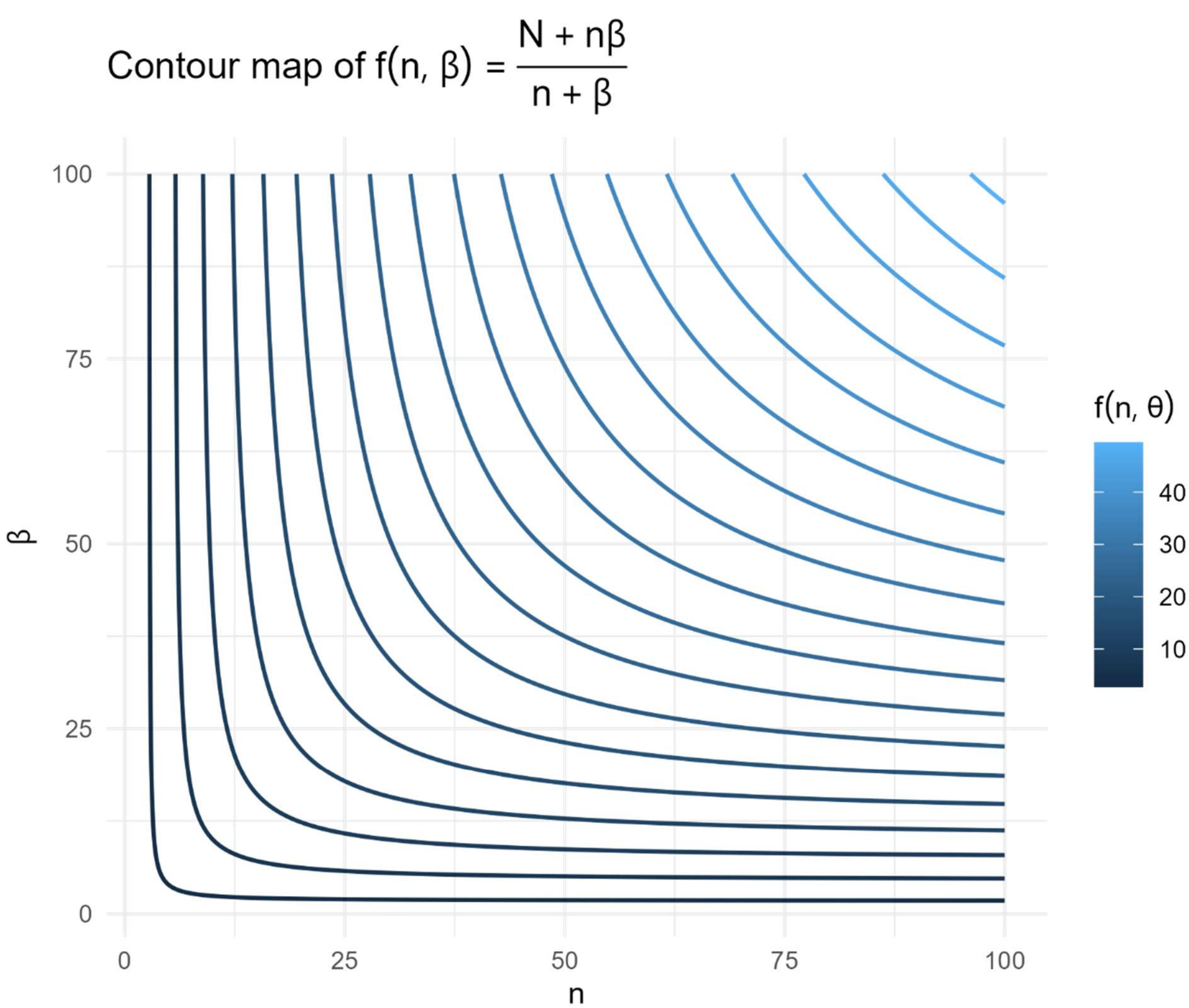
The effective sample size $n_{eff,t}$ in period t for the saturated Dirichlet Multinomial depends on β and $n_{obs,t}$

(1.16)
$$n_{eff,t} = \frac{n_{obs,t} + \beta n_{obs,t}}{n_{obs,t} + \beta}$$

And for a set of periods n_{eff} is

(1.17)
$$n_{eff} = \sum_{t=1}^{T} n_{eff,t} = \sum_{t=1}^{T} \frac{n_{obs,t} + \beta n_{obs,t}}{n_{obs,t} + \beta}$$

Here is a contour plot of the effective sample size function for the saturated Dirichlet Multinomial.



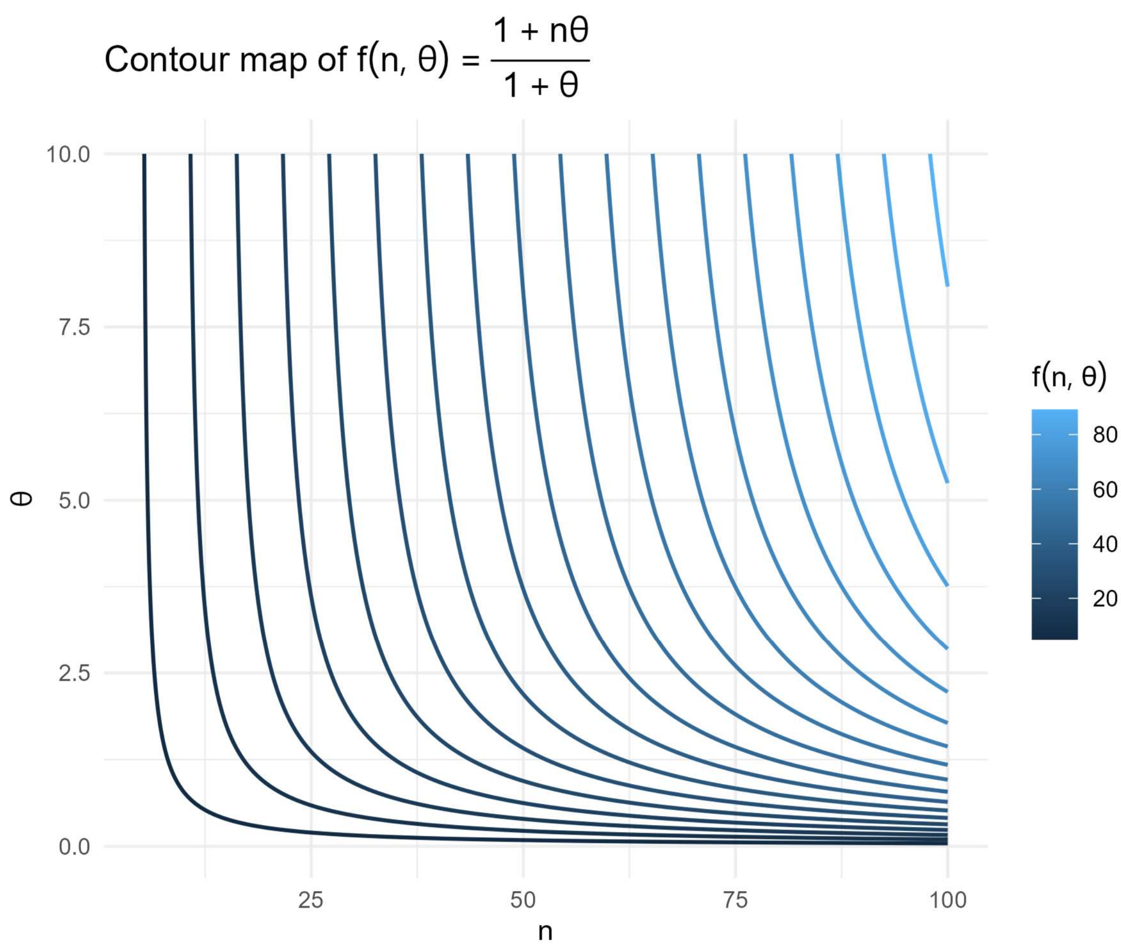
The effective sample size $n_{eff,t}$ in period t for the linear Dirichlet Multinomial is

(1.18) $$n_{eff,t} = \frac{1+\theta n_{obs,t}}{1+\theta}$$

And for a set of T periods n_{eff} is

(1.19) $$n_{eff} = \sum_{t=1}^{T} n_{eff,t} = \sum_{t=1}^{T} \frac{1+\theta n_{obs,t}}{1+\theta}$$

Here is a contour plot of the effective sample size function for the linear Dirichlet Multinomial.



We want to identify any critical points of the two Dirichlet Multinomial forms to help understand the curvature of the effective sample size function. The gradient for the effective sample size $\mathrm{n}_{eff,t}$ of the linear Dirichlet Multinomial is

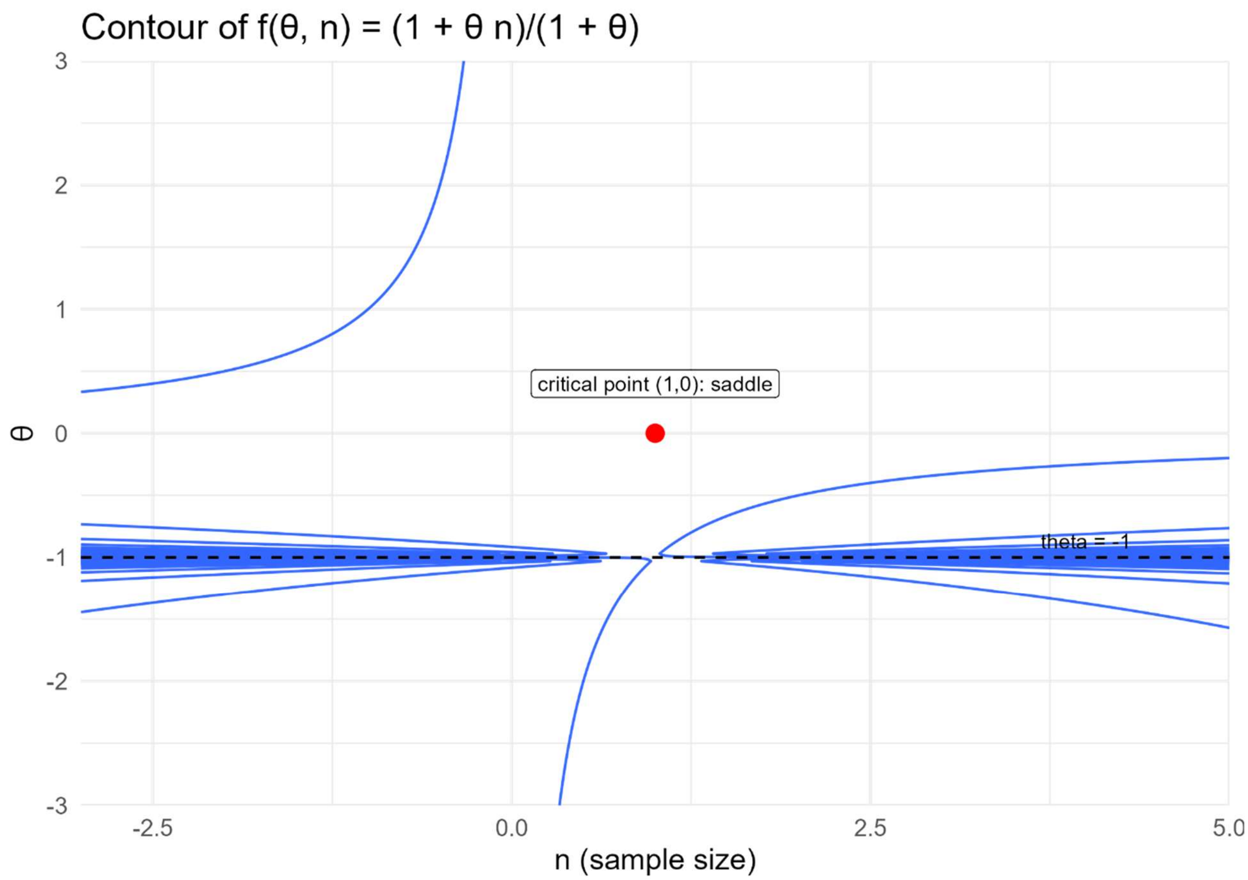
(1.20) $$\frac{\partial \mathrm{n}_{eff,t}}{\partial \mathrm{n}_{obs,t}} = \frac{\theta}{1+\theta} = 0 \Rightarrow \theta = 0 \ and \ \frac{\partial \mathrm{n}_{eff,t}}{\partial \theta} = \frac{\mathrm{n}_{obs,t} - 1}{\left(1+\theta\right)^2} = 0 \Rightarrow \mathrm{n}_{obs,t} = 1$$

There is a single critical point for the effective sample size at $\left(\mathrm{n}_{obs}, \theta\right) = \left(1, 0\right)$. The Hessian matrix for the effective sample size n_{eff} of the linear Dirichlet Multinomial evaluated at $\left(1,0\right)$ is

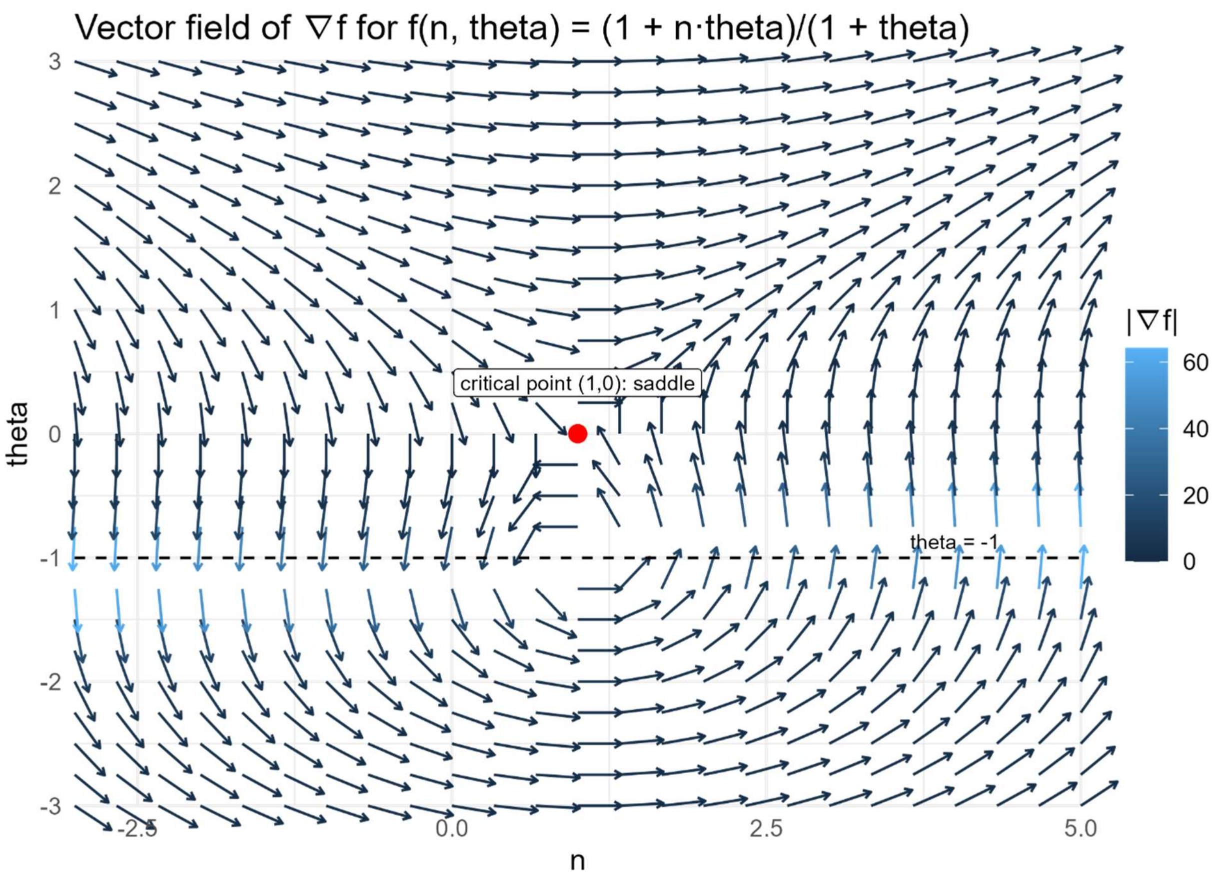
(1.21) $$H\left(0,1\right) = \begin{bmatrix} \dfrac{\partial^2 \mathrm{n}_{eff,t}}{\partial \mathrm{n}_t^2} & \dfrac{\partial^2 \mathrm{n}_{eff,t}}{\partial \mathrm{n} \partial \theta} \\ \dfrac{\partial^2 \mathrm{n}_{eff,t}}{\partial \mathrm{n} \partial \theta} & \dfrac{\partial^2 \mathrm{n}_{eff,t}}{\partial \theta_t^2} \end{bmatrix} = \begin{bmatrix} 0 & \dfrac{1}{\left(1+\theta\right)^2} \\ \dfrac{1}{\left(1+\theta\right)^2} & \dfrac{-2(\mathrm{n}_{obs,t} - 1)}{\left(1+\theta\right)^3} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Eigenvalues of H are $\lambda = \pm 1$ so $\left(\mathrm{n}_{obs}, \theta\right) = \left(1, 0\right)$ is a saddle point for the effective sample size $n_{eff,t}$ of the linear Dirichlet Multinomial.

Here is a plot showing the saddlepoint.



Here is a plot of the gradient vector field for the effective sample size function for the saturated Dirichlet Multinomial.



Similarly, the gradient for the effective sample size $n_{eff,t}$ of the saturated Dirichlet Multinomial is

$$(1.22)\ \frac{\partial \mathrm{n}_{eff,t}}{\partial \mathrm{n}_{obs,t}} = \frac{\beta(1+\beta)}{\left(\mathrm{n}_{obs,t}+\beta\right)^2} = 0 \ \Rightarrow\ \beta \in \{0,-1\}\ and\ \frac{\partial \mathrm{n}_{eff,t}}{\partial \beta} = \frac{\mathrm{n}_{obs,t}\left(\mathrm{n}_{obs,t}-1\right)}{\left(\mathrm{n}_{obs,t}+\beta\right)^2} = 0 \ \Rightarrow\ \mathrm{n}_{obs,t} \in \{0,1\}$$

and the two feasible critical points are $\left(\mathrm{n}_{obs,t},\beta\right)=(1,0)\ and\ \left(\mathrm{n}_{obs,t},\beta\right)=(0,-1)$.

The Hessian matrix for the effective sample size n_{eff} of the saturated Dirichlet Multinomial is

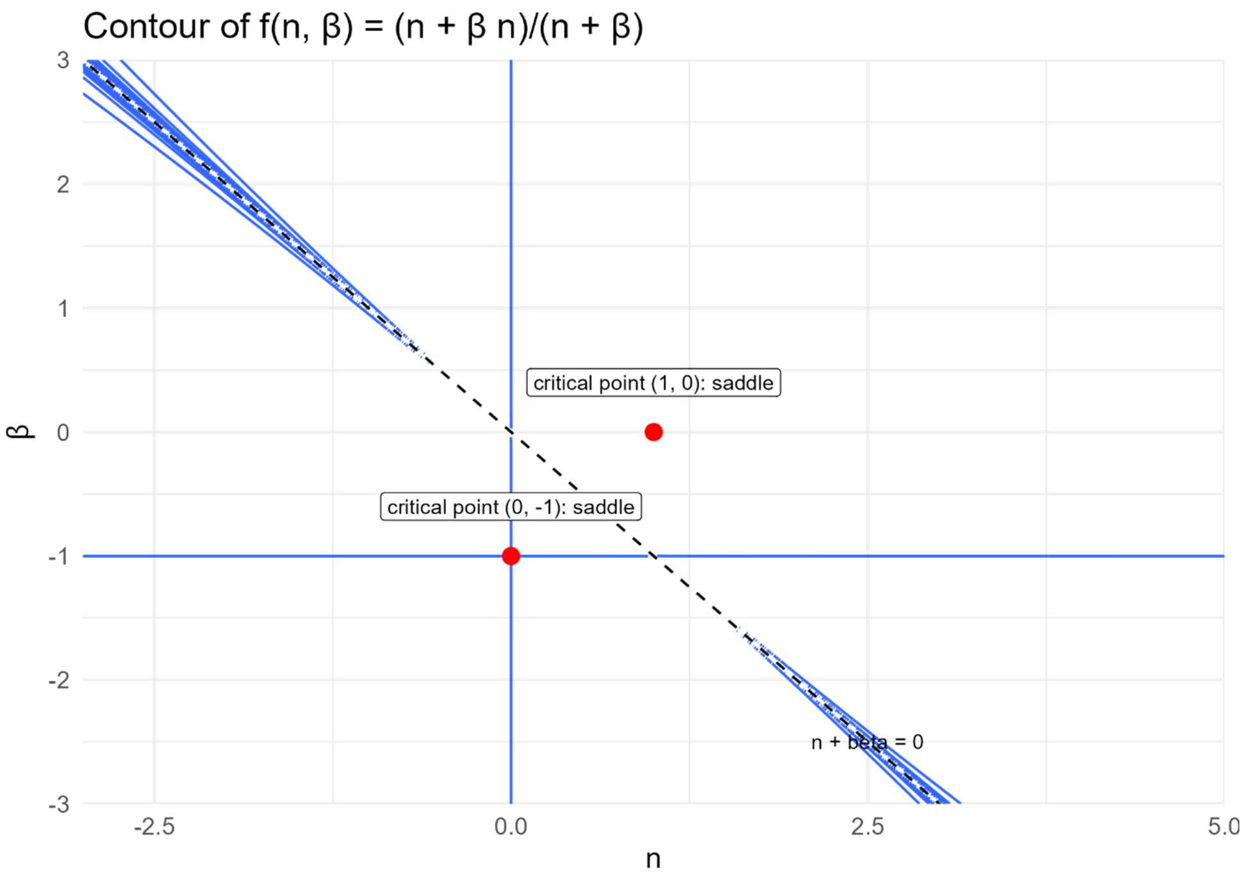
$$(1.23)\qquad H = \begin{bmatrix} \dfrac{\partial^2 \mathrm{n}_{eff,t}}{\partial \mathrm{n}^2_{obs,t}} & \dfrac{\partial^2 \mathrm{n}_{eff,t}}{\partial \mathrm{n}_{obs,t}\partial\beta} \\ \dfrac{\partial^2 \mathrm{n}_{eff,t}}{\partial \mathrm{n}_{obs,t}\partial\beta} & \dfrac{\partial^2 \mathrm{n}_{eff,t}}{\partial \beta_t^2} \end{bmatrix} = \begin{bmatrix} \dfrac{-2\left(\beta^2+\beta\right)}{\left(\mathrm{n}_{obs,t}+\beta\right)^3} & \dfrac{2\mathrm{n}_{obs,t}\beta+\mathrm{n}_{obs,t}-\beta}{\left(\mathrm{n}_{obs,t}+\beta\right)^3} \\ \dfrac{2\mathrm{n}_{obs,t}\beta+\mathrm{n}_{obs,t}-\beta}{\left(\mathrm{n}_{obs,t}+\beta\right)^3} & \dfrac{-2\mathrm{n}_{obs,t}{}^2+2\mathrm{n}_{obs,t}}{\left(\mathrm{n}_{obs,tt}+\beta\right)^3} \end{bmatrix}$$

The Hessians evaluated at the critical points $\left(\mathrm{n}_{obs,t},\beta\right)=\left(1,0\right)$ *and* $\left(\mathrm{n}_{obs,t},\beta\right)=\left(0,-1\right)$ are

(1.24)
$$H\left(1,0\right)=\begin{bmatrix}0&1\\1&0\end{bmatrix}\ \ and\ \ H\left(0,-1\right)=\begin{bmatrix}0&1\\1&0\end{bmatrix}$$

Eigenvalues of both Hessians are $\lambda=\pm 1$ so both $\left(\mathrm{n}_{obs,t},\beta\right)=\left(1,0\right)$ *and* $\left(\mathrm{n}_{obs,t},\beta\right)=\left(0,-1\right)$ are saddle points for the effective sample size $n_{eff,t}$ of the saturated Dirichlet Multinomial.

Here is a plot showing the saddlepoints.



Here is a plot of the gradient vector field for the effective sample size function for the saturated Dirichlet Multinomial.